

Customer Churn Prediction & LTV Engine

Week 4 • Day 2

Dashboard KPIs & Charts

Build the 6 key KPI charts for the Churn Analytics Dashboard

■ Day Objectives

- ◆ Create 6 KPI metrics: total customers, churn rate, avg LTV, high-risk count
- ◆ Build charts: churn by contract, churn by internet service, LTV distribution
- ◆ Create a customer risk score table with filtering
- ◆ Organise charts on a dashboard canvas
- ◆ Save and share the dashboard

■ 1. KPI Design Strategy

A dashboard tells a story at a glance. Our churn dashboard answers three questions for business stakeholders:

- **Who is churning?** Contract type, internet service, senior citizens
- **Who is at risk?** Current customers with high churn probability
- **Who is most valuable?** LTV segments and revenue at risk

■ 2. Create KPI Charts in Metabase

Chart 1: Total Customers KPI

New Question → Simple Question → customers_ltv → Count of Rows. Save as 'Total Customers'.

Chart 2: Churn Rate %

New Question → Native Query → paste this SQL:

■ Churn Rate SQL

```
SELECT
  ROUND(100.0 * SUM(CASE WHEN churn = 1 THEN 1 ELSE 0 END) / COUNT(*), 1) AS churn_rate_
pct
FROM customers_clean;
```

Chart 3: Average LTV

■ Avg LTV SQL

```
SELECT ROUND(AVG(total_ltv)::numeric, 2) AS avg_total_ltv
FROM customers_ltv;
```

Chart 4: High-Risk Customer Count

■ High Risk Count SQL

```
SELECT COUNT(*) AS high_risk_customers
FROM customers_ltv
WHERE churn_probability > 0.7;
```

Chart 5: Churn Rate by Contract Type — Bar Chart

■ Churn by Contract SQL

```
SELECT contract,
  COUNT(*) AS total,
  SUM(churn) AS churned,
  ROUND(100.0 * SUM(churn) / COUNT(*), 1) AS churn_rate
FROM customers_clean
GROUP BY contract
ORDER BY churn_rate DESC;
```

Chart 6: LTV Segment Distribution — Donut Chart

■ LTV Segment SQL

```
SELECT ltv_segment, COUNT(*) AS customer_count,
  ROUND(AVG(total_ltv)::numeric, 2) AS avg_ltv
FROM customers_ltv
GROUP BY ltv_segment;
```

■ 3. Commit & Checklist

- 6 KPI charts created in Metabase
- Dashboard canvas created with all 6 charts arranged
- Screenshot of dashboard saved to reports/figures/dashboard_screenshot.png

■ Week4-Day2 Commit

```
git add .  
git commit -m "Week4-Day2: Metabase KPI dashboard – 6 charts, churn + LTV visualisations"  
git push origin main
```

■ **Estimated Time: 5 – 6 hours**

— End of Week 4 Day 2 — Zaalima Development Confidential —